

1. Question

Write a Java program to check whether the given number is a Prime Palindrome or not.

A number is a Prime Palindrome if it is both:

Prime — it has exactly two factors: 1 and itself.

Palindrome — it remains the same when its digits are reversed.

Example: 131 is a Prime Palindrome.

2. Steps for Execution

1. Install Java JDK on the system.

2. Open a Java IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code.

3. Create a Java file named:

4. PrimePalindrome.java

5. Enter the given Java program.

6. Save the program.

7. Compile the program:

```
javac PrimePalindrome.java
```

8. Run the program:

```
java PrimePalindrome
```

9. Enter an integer when prompted.

10. The program reverses the number to check whether it is a palindrome.

11. The program counts the factors of the number to check whether it is prime.

12. If both conditions are satisfied, it displays Prime Palindrome.

Otherwise, it displays Not a Prime Palindrome.

3. Algorithm

Step 1: Start the program.

Step 2: Read the number n from the user.

Step 3: Store the original number in temp.

Step 4: Reverse the number using the following process:

$rev = rev \times 10 + temp \% 10$

$temp = temp / 10$

Repeat until temp becomes 0.

Step 5: Compare rev with the original number n.

- If $rev == n$, the number is a palindrome.
- Otherwise, it is not a palindrome.

Step 6: Initialize count = 0.

Step 7: Check every number from 1 to n.

If:

$n \% i == 0$

increment count.

Step 8: If count == 2, the number is prime.

Step 9: Check both conditions:

$rev == n \ \&\& \ count == 2$

Step 10: If both are true, display:

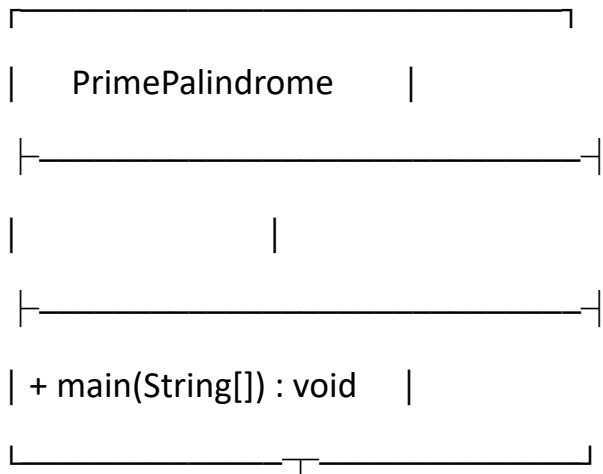
Prime Palindrome

Otherwise, display:

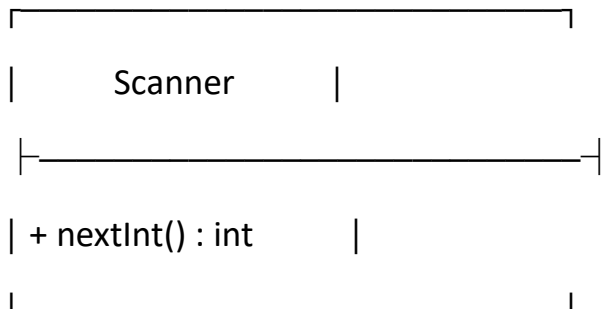
Not a Prime Palindrome

Step 11: Stop the program.

UML CLASS DIAGRAM:



| uses



CODE:

```
package javacore;
```

```
import java.util.Scanner;
```

```
public class PrimePalindrome {
```

```
    public static void main(String[] args) {
```

```
        Scanner ac = new Scanner(System.in);
```

```
        System.out.print("Enter a number to check primepalindrome or not: ");
```

```
        int n = ac.nextInt();
```

```
        int rev = 0, temp = n, count = 0;
```

```
while (temp > 0) {  
    rev = rev * 10 + temp % 10;  
    temp = temp / 10;  
}  
for (int i = 1; i <= n; i++) {  
    if (n % i == 0)  
        count++;  
}  
if (rev == n && count == 2)  
    System.out.println(n + " is a Prime Palindrome");  
else  
    System.out.println(n + " is not a Prime Palindrome");  
}  
}
```

OUTPUT SCREENSHOT (SAMPLE RUN):

Example 1 — Prime Palindrome

Input:

Enter a number to check primepalindrome or not: 131

Processing:

Reverse of 131 = 131

Number of factors = 2

Output:

131 is a Prime Palindrome